Stigler's Diet Problem

Solution with OpenRules Rule Solver (without Java)

The problem is offered as a Decision Management Community Challenge October-2026:

> *Stigler's Diet Problem is one of the classical problems in Operational Research. The problem finds its origin in the work of Nobel laureate George Stigler (1944), who posed this as a mathematical exercise, not as eating recommendations:*
>
> *For a moderately active man weighing 154 pounds, how much of each of 77 foods should be eaten on a daily basis so that the man's intake of nine nutrients will be at least equal to the recommended dietary allowances in 1943, with the cost of the diet being minimal?*
>
> *You can find more details, the nutrients and commodities lists, along with a possible solution, at the OR-Tools website.*

Glossary

Glossary glossary				
Variables	**Business Concept**	**Attribute**	**Type**	**Used As**
Nutrients	**DietProblem**	nutrients	Nutrient[]	in
Foods		foods	Food[]	in
TotalAnnualCost		totalAnnualCost	double	out
NutrientName	**Nutrient**	name	String	in
NutrientId		id	int	in
DailyIntake		dailyIntake	double	in
FoodId	**Food**	foodId	String	in
Commodity		commodity	String	in
Units		units	String	in
Price		price	double	in
NutrientQuantities		nutrientQuantities	double[]	in
AnnualCost		annualCost	double	out
NumberOfNutrients	:=	${Nutrients}.length	int	
NutrientQuantity		${NutrientQuantities}[${NutrientId}-1]	double	

Test Data

We copied the OR-Tools data to Excel and added two lines to the headers.

DecisionData Nutrient nutrients		
NutrientId	NutrientName	DailyIntake
1	Calories (calories)	3
2	Protein (grams)	70
3	Calcium (grams)	0.8
4	Iron (milligrams)	12
5	Vitamin A (IU)	5
6	Thiamine (Vitamin B1) (milligrams)	1.8
7	Riboflavin (Vitamin B2) (milligrams)	2.7
8	Niacin (milligrams)	18
9	Ascorbic Acid (Vitamin C) (milligrams)	75

DecisionData Food foods												
FoodId	**Commodity**	**Units**	**Price**	**NutrientQuantities**								
F1	Wheat Flour (Enriched)	10 lb.	36	45	1411	2	365	0	55	33	441	0
F2	Macaroni	1 lb.	14.1	12	418	0.7	54	0	3.2	1.9	68	0
F3	Wheat Cereal (Enriched)	28 oz.	24.2	12	377	14	175	0	14	8.8	114	0
F4	Corn Flakes	8 oz.	7.1	11	252	0.1	56	0	14	2.3	68	0
F5	Corn Meal	1 lb.	4.6	36	897	1.7	99	30.9	17	7.9	106	0
F6	Hominy Grits	24 oz.	8.5	29	680	0.8	80	0	11	1.6	110	0
F7	Rice	1 lb.	7.5	21	460	0.6	41	0	2	4.8	60	0
F8	Rolled Oats	1 lb.	7.1	25	907	5.1	341	0	37	8.9	64	0
F9	White Bread (Enriched)	1 lb.	7.9	15	488	2.5	115	0	14	8.5	126	0
F10	Whole Wheat Bread	1 lb.	9.1	12	484	2.7	125	0	14	6.4	160	0
F11	Rye Bread	1 lb.	9.1	12	439	1.1	82	0	9.9	3	66	0
F12	Pound Cake	1 lb.	24.8	8	130	0.4	31	18.9	2.8	3	17	0
F13	Soda Crackers	1 lb.	15.1	13	288	0.5	50	0	0	0	0	0
F14	Milk	1 qt.	11	6.1	310	11	18	16.8	4	16	7	177
F15	Evaporated Milk (can)	14.5 oz.	6.7	8.4	422	15	9	26	3	24	11	60
F16	Butter	1 lb.	30.8	11	9	0.2	3	44.2	0	0.2	2	0
F17	Oleomargarine	1 lb.	16.1	21	17	0.6	6	55.8	0.2	0	0	0

...

DecisionTest testCases		
#	Define	Define
Test ID	**Nutrients**	**Foods**
1	nutrients	foods

Business Logic

We implemented using the newest version of Rule Solver without any Java code using only OpenRules decision tables.

Problem Definition

We define our problem in three steps described in the table "Define":

Decision Define
Execute
Decision Tables
CreateVariables
PostConstraints
SetObjective

Step 1. Create Main Constraint Variables

Decision **CreateVariables** [for each Food in Foods]		
SolverDefine		
Variable Name	**Domain**	**Array Name**
{{FoodId}}	0.0..1000000.0	"AllFoodVariables"

For every food in the array "Foods" this table creates unknown decision variables defined from 0 to a large number (we selected 1,000,000) and adds it to the array "AllFoodVariables" that will be used to define the objective "TotalAnnualCost".

Step 2. Post Daily Intake Constraints

Decision **PostConstraints** [for each Nutrient in Nutrients]
Execute
Tables
CreateConsumedNutrientArray
PostDailyIntakeConstraints

For each nutrient in the array "Nutrients" this table will create an array of constrained variables "Consumed{{NutrientId}}" and then post a constraint

*Sum (Consumed{{NutrientId}}) >= ({{DailyIntake of Nutrient}} * 365.0)*

Decision **CreateConsumedNutrientArray** [for each Food in Foods]		
SolverDefine		
Variable Name	**Variable Expression**	**Array Name**
Consumed_{{NutrientId}}_{{FoodId}}	{{FoodId}} * {{NutrientQuantity}}	Consumed{{NutrientId}}

Decision **PostDailyIntakeConstraints**
SolverPost
Variable Name
Sum (Consumed{{NutrientId}}) >= ({{DailyIntake of Nutrient}} * 365.0)

Step 3. Set Optimization Objective

Decision SetObjective		
SolverDefine		**SolverSetObjective**
Variable Name	**Expression**	**Objecive**
"TotalAnnualCost"	Sum("AllFoodVariables")	"TotalAnnualCost"

This table defines a constrained variable "*TotalAnnualCost*" as *Sum("AllFoodVariables")* and sets it as the optimization objective.

Problem Resolution

We use the predefined method "SolverMinimize" calling it from the standard table "Solve":

Decision **Solve**
Execute
Actions
SolverMinimize
AssignSolution

After an optimal solution is found, this table will invoke the following table to assign the found results to the output decision variables "AnnualCost" for each food in the array "Foods":

Decision **AssignSolution** [for each Food in Foods]					
SolverIfSolution			**SolverAssignSolution**		**Print**
Variable Name	**Oper**	**Value**	**Business Variable Name**	**Solver Variable Name**	**Text**
{{FoodId}}	>	0	"AnnualCost"	{{FoodId}}	{{Commodity}} = $ {{AnnualCost}}

Applied Linear Solvers

We tried different linear/MIP solvers by changing the property “SOLVER” inside the “test.bat”:

```
@echo off
cd %~dp0
set SOLVER=Scip
rem set SOLVER=CLP
rem set SOLVER=ojAlgo
call mvn clean test -e
```

They all produced good results:

```
TotalAnnualCost[39.661732224619 47]
*** Execution Profile ***
Execution time: 1089 msec
SolverSaveSolution
Wheat Flour (Enriched) = $ 10.7744567645
Liver (Beef) = $ 0.6907836618
Cabbage = $ 4.0932688874
Spinach = $ 1.8277959785
Navy Beans, Dried = $ 22.2754269323
'testCases-1' succeeded. Elapsed time 1590.03 ms
```